

Evolution operators for describing linearly singular systems

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The Picard–Lindelöf theorem guarantees the **existence and uniqueness** of solutions for an ODE in normal form:

$$\dot{x}^i = f^i(x),$$

provided that the functions f^i are sufficiently regular.

Geometrically, these systems correspond to **vector fields**:

- If X is a v.f., in local coordinates it can be written as:

$$X = \sum_{i=1}^n f^i(x) \frac{\partial}{\partial x^i}.$$

- The integral paths of X , i.e., paths $\gamma: I \rightarrow M$ satisfying $\dot{\gamma} = X \circ \gamma$, are precisely the solutions of the ODE $\dot{x}^i = f^i(x)$.

Introduction

However, not all ODEs can be written in normal form ($\dot{x} = f(x)$).

- **Non-holonomic mechanics.**
- **Singular Lagrangians.**
- **Electrical circuit design** (differential-algebraic equations / DAEs).
- **Control of interconnected systems.**

Example (Euler–Lagrange equations)

The Euler–Lagrange equations for a Lagrangian $L(q, \dot{q})$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = 0 \quad \Longleftrightarrow \quad \frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j} \ddot{q}^j = \frac{\partial L}{\partial q^i} - \frac{\partial^2 L}{\partial \dot{q}^i \partial q^j} \dot{q}^j.$$

can only be written in normal form if the Lagrangian is **regular** (i.e., if we can invert the Hessian matrix $\frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j}$).

Differential relations

A general autonomous first-order ODE

$$f^i(x, \dot{x}) = 0$$

can be seen, geometrically, as a subset (or submanifold) $D \subseteq TM$.

A path $\xi: I \rightarrow M$ is a **solution** if its tangent lift lies entirely within D :

$$\dot{\xi}(I) \subset D.$$

Remark

If $D \subset TM$ is a submanifold, then we can regard it as being implicitly defined, locally, as the zero level set of some functions $\{f^i = 0\}$. Thus, locally, a path $(x(t), \dot{x}(t))$ is a solution of $D \subset TM$ iff it solves the ODE

$$f^i(x, \dot{x}) = 0.$$

If the ODE is in normal form, $D = X(M)$ for some v.f. $X \in \mathfrak{X}(M)$.

Differential relations

Definition

We say that a subset $D \subseteq TM$ is a **differential relation**.

For a general differential relation, we **might not have solutions** for every initial condition, and even when they exist they **might not be unique**.

Example

Let $M = \mathbb{R}$ and define the differential relation

$$D = \{(x, v) \in T\mathbb{R} \mid v^2 = x\}.$$

For the initial condition $p_0 = 0 \in D$ the constant path $\xi_1(t) = 0$ and $\xi_2(t) = \frac{1}{4}t^2$ are solutions, so there is no uniqueness of solutions.

However, there is no solution for any initial condition $p = x$ with $x < 0$.

Constraint algorithm

In order to study the existence of solutions in this more general setting, one develops the **constraint algorithm**. We want to find $D' \subset TM$ s.t.:

- 1 It is *equivalent* to the original differential relation, so D and D' have the **same solution paths**, and
- 2 There exists a solution through every initial condition $p_0 \in \tau_M(D')$.

Constraint algorithm

Any solution $\xi(t)$ of $D \subset TM$ must lie in the projected base space and remain tangent to it at all times. This leads to the iterative method:

$$M_{i+1} := \tau_M(D_i), \quad D_{i+1} := D_i \cap TM_{i+1},$$

starting with $D_0 = D$.

At each step, we have to assume that M_{i+1} is a closed submanifold, and the algorithm will finish at a **final constraint submanifold** M_f .

Linearly singular systems: Initial definitions

We restrict our analysis to a more tractable subcase that models most physical systems of interest (e.g., the Euler–Lagrange equations), where the implicit system of ODEs is of the form

$$A(x)\dot{x} = b(x),$$

with A a, generally singular, matrix and b a vector. Geometrically,

Definition (Linearly singular system)

Let M be a manifold and $\pi: F \rightarrow M$ a vector bundle.

A **linearly singular system** (or lss) is a pair (A, b) where:

- $A: TM \rightarrow F$ is a morphism of vector M -bundles,
- $b: M \rightarrow F$ is a section of π .

A path $\xi: I \rightarrow M$ is a **solution** of (A, b) if we have $A \circ \dot{\xi} = b \circ \xi$.

Linearly singular systems: Initial definitions

In terms of commutative diagrams,

$$\begin{array}{ccc} TM & \xrightarrow{A} & F \\ \tau_M \downarrow & \nearrow \pi & \\ M & & \end{array} \quad \begin{array}{c} \nearrow b \end{array}$$

$$\begin{array}{ccccc} & & TM & \xrightarrow{A} & F \\ & \nearrow \dot{\xi} & \downarrow \tau_M & \nearrow \pi & \\ I & \xrightarrow{\xi} & M & & \end{array} \quad \begin{array}{c} \nearrow b \end{array}$$

In terms of differential relations, a linearly singular system defines the subset

$$D = A^{-1}(b(M)) \subset TM.$$

Note that this might not be a submanifold.

Explicit realization of the constraint algorithm

Consider a linearly singular system (A, b) . We want to find the set of points for which we have existence of solutions.

Since A is not an isomorphism in general, solutions $X(x) \in T_x M$ of

$$A_x \cdot X(x) = b(x)$$

may not exist at every point $x \in M$.

Consistency condition: Solutions can only exist where $b(x)$ has a valid preimage. This defines the *first constraint submanifold*:

$$M_1 := \{x \in M \mid b(x) \in \text{Im } A_x\}$$

If A has locally constant rank,

$$b(x) \in \text{Im } A_x \quad \Longleftrightarrow \quad b(x) \in (\text{Ker } {}^t A_x)^\circ$$

Explicit realization of the constraint algorithm

So, if $\{s^i\}$ is a local frame of $\text{Ker } {}^tA$, then M_1 is described by

$$\phi^i = \langle s^i, b \rangle \quad \text{and} \quad M_1 = \{\phi^i = 0\}.$$

Definition

A function ϕ is a **constraint function** if $\phi \circ \xi = 0$ for any solution path ξ .

The constraint algorithm then continues (assuming constant rank and closed submanifold conditions) by applying the tangency condition

$$(X \cdot \phi^i) = 0,$$

where X is a vector field s.t. $A \circ X = b$ is satisfied at points of M_1 . This either yields new constraints or fixes some degrees of freedom. Applying this iteratively gives the constraints defining the final constraint submanifold.

Example: Hamiltonian systems

Definition

A **Hamiltonian system** is a triple (M, ω, H) , where M is a manifold, $\omega \in \Omega^2(M)$ is a closed 2-form of constant rank and H is a function.

The solutions to Hamilton's equations are

$$\widehat{\omega} \circ \dot{\xi} = dH \circ \xi,$$

where $\widehat{\omega}: TM \rightarrow T^*M$ is the vector bundle morphism induced by ω . This is also the equation of motion of the linearly singular system $(\widehat{\omega}, dH)$

$$\begin{array}{ccc} TM & \xrightarrow{\widehat{\omega}} & T^*M \\ \tau_M \downarrow & \nearrow \pi_M & \\ M & \xleftarrow{dH} & \end{array}$$

Hamiltonian systems appear naturally in the study of the singular Lagrangians in classical mechanics. Geometrically, they were first studied by Gotay, Nester and Hinds in 1978.

Example: Dirac structures

A Dirac structure is a vector subbundle $L \subseteq TM \oplus T^*M$ s.t. $L = L^\perp$,

$$L^\perp = \{(Y, \beta) \in TM \oplus T^*M \mid \langle \alpha, Y \rangle + \langle \beta, X \rangle = 0 \text{ for all } (X, \alpha) \in L\}.$$

The dynamics associated with a Dirac structure are implicit Hamiltonian systems, i.e. given a function H we look for the vectors X such that

$$(X, dH) \in L$$

In terms of linearly singular systems, the integral paths of the vector fields X are the solutions to the linearly singular system

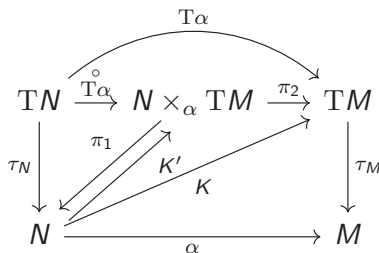
$$\begin{array}{ccc} TM & \xrightarrow{\iota} & (TM \oplus T^*M)/L \\ \downarrow & \nearrow & \\ M & & \end{array} \quad \begin{array}{c} \\ \\ \end{array} \quad \begin{array}{c} \\ \\ [(0, -dH)] \end{array}$$

Example: Vector fields along maps

A **vector field along a map** $\alpha: N \rightarrow M$ is a map $K: N \rightarrow TM$ such that:

$$\tau_M \circ K = \alpha.$$

Equivalently, a vector field along a map α is a section of the pullback bundle $N \times_{\alpha} TM$. With this, a vector field along a map defines a lss



where the maps are $\overset{\circ}{T}\alpha = (\tau_N, T\alpha)$ and $K' = (\text{Id}_N, K)$.

Examples: Vector fields along maps

- The solutions of this linearly singular system are given by the paths $\xi: I \rightarrow N$ such that

$$T\alpha \circ \dot{\xi} = K \circ \xi.$$

- If the path ξ is a solution of the system then, for any function $h \in \mathcal{C}^\infty(M)$, we have

$$\frac{d}{dt}(\alpha^*(h) \circ \xi) = (K \cdot h) \circ \xi.$$

- Any vector field along a map naturally defines another differential relation (not necessarily a lss), given by

$$K(N) \subseteq TM.$$

Description as a vector field along a map

Aim of this work

Determine which linearly singular systems admit a **description in terms of a vector field along a map**.

This is advantageous because:

- A vector field along a map naturally defines **two differential relations**

$$D = T\alpha^{-1}(K(N)) \subseteq TN \quad \text{and} \quad D' = K(N) \subseteq TM.$$

which can be seen as a “Lagrangian” and a “Hamiltonian” formalism.

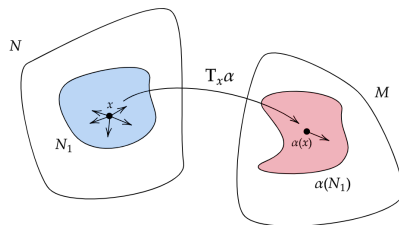
- Under suitable conditions, K can be used to **relate the solutions and constraint functions** of both systems.
- This formulation (known as the description in terms of an **evolution operator**) is useful for dealing with singular Lagrangian systems in both classical and contact mechanics.

Differential relation as a vector field along a map

Let $D \subset TN$ and $\alpha: N \rightarrow M$, s.t. $N_1 = \tau_N(D)$ is a closed submanifold. If, for any $x \in N_1$ and $v_1, v_2 \in D_x$, we have

$$T_x\alpha(v_1) = T_x\alpha(v_2).$$

Then $T_x\alpha$ **collapses** each fibre D_x **to a single vector** in $T_{\alpha(x)}M$.



So there exists a unique map (not necessarily smooth) $K: N_1 \rightarrow TM$ along $\alpha|_{N_1}$ such that $T_x\alpha(D_x) = K(x)$ for all $x \in N_1$.

Differential relation as a vector field along a map

This vector field along a map defines (under regularity assumptions) a linearly singular system, whose solutions are the paths $\xi: I \rightarrow N$ s.t.:

$$T\alpha \circ \dot{\xi} = K \circ \xi.$$

Thus,

- We have three differential relations:

$$D \subset TN, \quad T\alpha^{-1}(K(N_1)) \subseteq TN, \quad T\alpha(D) = K(N_1) \subseteq TM,$$

- Solutions of the first are solutions of the second. The converse is not true in general.
- Solutions of the first and second map to solutions of the third by composing with α . But, again, not every solution has this form.

Differential relation as a vector field along a map

Let $K: \mathbb{R}^2 \rightarrow T\mathbb{R}$ be given by $K(x, y) = (x, xy)$ over $\alpha(x, y) = x$.

We have

$$K(\mathbb{R}^2) = \{(x, v) \in T\mathbb{R} \mid x \neq 0 \text{ or } v = 0\},$$

Thus, $\eta(t) = t^2$ satisfies $\dot{\eta}(I) \subset K(\mathbb{R}^2)$.

However, solving $K(\xi(t)) = (\eta(t), \dot{\eta}(t))$ for a path $\xi(t) = (x(t), y(t))$ forces $x(t) = t^2$ and $y(t) = 2/t$ for $t \neq 0$. As $y(t)$ tends to infinity as $t \rightarrow 0$, the path $\xi(t)$ cannot be continuous.

Proposition

Let K be a vector field along a map α with **locally constant rank**, and let $\eta: I \rightarrow M$ be a solution of the differential relation $K(N)$. Then,

- There locally exists a smooth path $\xi: I \rightarrow N$ s.t. $\dot{\eta} = K \circ \xi$.
- If, moreover, K is an injective immersion, ξ is unique.
- If, moreover, K is an embedding, then ξ is unique and exists globally.

Linearly singular systems as vector fields along maps

Now, assume that D is described by a linearly singular system (A, b) , so $D = A^{-1}(b(N))$. In this case,

$$T_x \alpha(v_1 - v_2) = 0, \iff \text{Ker } A_x \subseteq \text{Ker } T_x \alpha,$$

for any $v_1, v_2 \in D_x$ and $x \in N_1$.

If A has locally constant rank, this condition implies the existence of a **unique vector bundle map**

$$\psi: \text{Im } A|_{N_1} \rightarrow TM,$$

such that $\psi \circ A|_{N_1} = T\alpha|_{N_1}$.

With this map, we can write the **vector field along** α explicitly, as

$$K = \psi \circ b|_{N_1}.$$

Linearly singular systems as vector fields along maps

If, moreover, $\text{Ker } A = \text{Ker } T\alpha$, then ψ defines an **isomorphism** between $\text{Im } A_x$ and $\text{Im } T_x\alpha$. And, in this case,

$$\xi \text{ solves } (A, b) \iff \xi \text{ solves the system defined by } K.$$

i.e.

$$A \circ \dot{\xi} = b \circ \xi \iff T\alpha \circ \dot{\xi} = K \circ \xi.$$

In other words,

Theorem

Let (A, b) be a linearly singular system on N . If there exists a map $\alpha: N \rightarrow M$ such that

$$\text{Ker } A = \text{Ker } T\alpha,$$

then the system (A, b) can be described in terms of a vector field along α .

Example: The evolution operator for classical mechanics

In classical mechanics, from a Lagrangian function $L \in \mathcal{C}^\infty(TQ)$, one can define the following linearly singular system:

$$\begin{array}{ccc} TTQ & \xrightarrow{(J, \widehat{\omega}_L)} & TTQ \oplus T^*TQ \\ \downarrow & \nearrow (\Delta, dE_L) & \\ TQ & & \end{array}$$

where J is the vertical endomorphism, ω_L is the Lagrangian 2-form, Δ is the Liouville vector field, and $E_L = \Delta(L) - L$ is the Lagrangian energy.

The solutions $\xi: I \rightarrow TQ$ to this linearly singular system

$$\widehat{\omega}_L \circ \dot{\xi} = dE_L \circ \xi, \quad J \circ \dot{\xi} = \Delta \circ \xi,$$

are the solutions to the Euler–Lagrange equations. The second equation is just the SODE condition.

Example: The evolution operator for classical mechanics

Now, if one considers the Legendre map

$$\begin{aligned} \mathcal{F}L: \quad TQ &\longrightarrow T^*Q \\ (q, v) &\longmapsto \left(q, \frac{\partial L}{\partial v}(q, v) \right) \end{aligned}$$

Then one can prove that

$$\text{Ker}(J, \widehat{\omega}_L) = \text{Ker } J \cap \text{Ker } \widehat{\omega}_L = \text{Ker } T\mathcal{F}L,$$

and, hence, there exists a **vector field along the Legendre map**, $K: TQ \rightarrow TT^*Q$, which provides a description of the Euler–Lagrange equations. In coordinates,

$$K(q, v) = v^i \frac{\partial}{\partial q^i} \Big|_{\mathcal{F}L(q, v)} + \frac{\partial L}{\partial q^i} \frac{\partial}{\partial p_i} \Big|_{\mathcal{F}L(q, v)}.$$

- We have identified the **conditions** under which a linearly singular system can be described with an **evolution operator**.
- This can be successfully **applied to recover** the vector fields along the Legendre map $\mathcal{FL}$ of both **classical mechanics and contact mechanics**.

What's next?

- Apply this formalism to **relate and classify** the **constraints** of Lagrangian and Hamiltonian formalisms,
- Find **descriptions** in terms of vector fields along maps for **other mechanical systems** (time-dependent, non-holonomic...).

- W. C. Rheinboldt, “Differential-algebraic systems as differential equations on manifolds”, *Math. Comput.* **43** (1984), 473–482.
- G. Marmo, G. Mendella, W. M. Tulczyjew, “Symmetries and constants of the motion for dynamics in implicit form”, *Ann. Inst. H. Poincaré Phys. Théor.* **57** (1992), 147–166.
- X. Gràcia and J. M. Pons, “A generalized geometric framework for constrained systems”, *Differ. Geom. Appl.* **2** (1992), 223–247.
- X. Gràcia and J. M. Pons, “Symmetries and infinitesimal symmetries of singular differential equations”, *J. Phys. A: Math. Gen.* **35** (2002), 5059–5077.
- X. Gràcia and J. M. Pons, “On an evolution operator connecting Lagrangian and Hamiltonian formalisms”, *Lett. Math. Phys.* **17** (1989), 175–180.